TRADING EDGE LAB / ADDENDUM 2.0

Adaptive Edge

A master algorithm for better decisions, faster learning and disciplined control

Audit, deeper research and an implementable operating method for a strategy, sleeve or trading desk.

THE DECISION THAT MATTERS

Which feasible action improves the portfolio under the evidence we have, before the opportunity expires - and is further analysis worth its cost?

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30 additional primary sources. Explicit equations, control logic, evidence contracts, operational tests and a small offline reference kernel. No market profitability or live deployment claimed.

1 / Audit and deeper findings

The original Trading Edge Lab succeeds at the difficult first task: separating a story about profitable trading from an economically plausible, executable and testable advantage. Its strongest requirements remain intact: identify who values what the desk provides; compare beliefs with executable prices; condition on fills; price joint bad-state exposure; preserve chronology; and account for strategy search.

The missing layer was an explicit adaptive controller. The first version required monitoring, reconciliation and evidence gates but left substantial freedom in how those components interact. That freedom can produce an impressive-looking system that retrains on selected outcomes, confuses historical evidence with current opportunity, or changes an order while forgetting exposure that can still fill.

The upgrade is a practical control-and-learning addendum. It makes awareness a set of observable states, intelligence a decision-quality objective, adaptability a tested update rule, and speed an economic deadline. It does not claim that a controller can make every strategy profitable or produce an orders-of-magnitude return improvement without measurement.

What was reviewed and changed

All five authored v1 Markdown files were reviewed: core instructions, foundations, ten mechanism blueprints, validation and the 50-entry source register. This was a review of their full content; it was not a fresh complete reading of every v1 paper. Four independent research tracks then examined control theory, adaptive inference, economics and system behavior. The new register adds 30 distinct primary sources, with access scope and limitations stated individually. One re-examined source is explicitly retained as an alias, not counted again.

Area	Finding in v1	Change in v2
Economic foundations	Main distinctions and displayed identities remain sound in this review	Preserve them; make the core arbitrage definition match the fuller initial-cost and positive-probability conditions
Dynamic decisions	Response choices named, control contract underspecified	Explicit state, budgets, feasible action menu and common-model portfolio comparison
Adaptation	Monitoring and refitting mentioned	Separate action response, approved learning and promotion of new rules
Evidence	Chronology and repeated testing recognized	Typed estimands, support checks, delayed labels, full information history and reset/error budgets
Awareness	Freshness/reconciliation intentions	Revision semantics, possible fills, atomic reservations and capability-specific degradation
Speed	Latency recognized as a cost	Deadline-aware computation and end-to-end tail measurement
Scope	Research framework	Research framework plus a small tested offline decision kernel, with explicit unimplemented runtime boundaries

The deepest additional insight

The scarce object is not a prediction. It is a **feasible action whose advantage survives uncertainty, competition, execution, capital constraints and delay**. This changes the system's priorities.

An uncertain forecast can be sufficient when every plausible value leads to the same action. A precise forecast can be useless at the offered price. A high-quality signal can lose its edge through selected fills. A good sleeve can reduce desk value by occupying collateral needed for a better opportunity. More information can arrive too late. An adaptive rule can improve its training score while changing its environment for the worse.

The master algorithm therefore allocates five budgets: capital, liquidity, time/compute, information acquisition and statistical error. It preserves each budget's meaning rather than combining unlike quantities into a confidence score. It also distinguishes three clocks: execution, belief updates and policy promotion.

Connections that change the design

Connection	Useful consequence	Boundary
Partial observability	Retain probabilities over market conditions and sets of possible own-account states	A regime label is not truth; a filter with full-sample fitted parameters leaks
Robust decision theory	Compare candidate and incumbent under the same plausible model	A scenario minimum is not a confidence interval
Rational metareasoning	Spend computation where it can change a valuable action before expiry	LLM certainty and longer reasoning are not measured information value
Sequential inference	Monitor continuously with a declared null and information history	Historical average evidence is not a current-edge forecast
Causal/off-policy evaluation	Learn from actions the data can actually support	Simulated fills cannot supply missing action probabilities or real executions
Performative prediction	Model how deployment and size change later observations	Stable retraining need not optimize deployed outcomes
Resource economics	Allocate shared funding, inventory and capacity by marginal portfolio value	Local shadow prices are not extra realized fees
Distributed systems	Preserve causal ordering, retry identity and uncertain order exposure	Local idempotency does not prove exactly-once broker execution

These connections are synthesized into operational rules in the following chapters. They are not claimed as novel academic discoveries. Several are deliberately conservative because the cost of a false action and the cost of a missed action differ by mandate; the chosen balance must be explicit and evaluated.

What the master algorithm actually does

At each eligible event it validates what the desk knows, reconciles what it might already own, updates approved beliefs, constructs a small action menu, evaluates full portfolio consequences, and performs only worthwhile additional computation. It chooses an admissible action or retains the feasible incumbent, reserves resources, and records a reproducible intent. It routes integrity failures to an independent contingency path.

When outcomes mature, it scores the decisions that were actually issued, separates prediction from capture and portfolio effects, and updates a prespecified evidence process. A new rule must outperform a frozen feasible baseline through an honest experiment before promotion. This prevents rapid learning from becoming rapid unrecorded search.

The most useful daily output may be an entry, a hold, a reduction, a hedge, a reconciliation request, a research experiment or no new exposure. The goal is useful coverage of opportunities and reliable management of existing risk, rather than a required number of trades.

How to implement this leanly

Begin with the finite approved candidate menu and one shared portfolio arbiter. Keep the event journal, state reducer, scoring, reservations and order manager modular but close together. Run research and evidence analysis asynchronously. Add sophisticated latent-state models, dynamic solvers or distributed services only after a simpler baseline reveals a measured limitation they can address.

The included reference kernel computes one deliberately small part: the paired robust improvement gate with supplied scenarios, evidence flags, freshness, deadlines and additive load constraints. It has no market feeds, broker integration, learning, durable event store or order submission. Its executable tests check mathematical and control invariants; they do not validate a trading edge. The runtime chapter specifies the additional failure tests required for a real adapter.

Use the updated skill with: “Use $engineer-trading-edge to audit this strategy, identify its binding loss of value, and apply the adaptive-control addendum. Preserve its approved timing authority, define the smallest useful controller, and run the next decisive experiment.”

The standalone prompt accompanying this report can be supplied to another LLM. The full skill remains preferable when its references and scripts are available: it loads deeper material only when needed and retains the original economic foundations.

2 / The master control algorithm

1. Objective and limits

Optimize a feasible decision using current evidence, executable payoffs, existing inventory, scarce resources, and the cost of changing course. Preserve the edge contract: identify what earns compensation before optimizing how to capture it. A control system cannot supply an absent economic advantage.

Elementary no-universal-edge result. In a finite-horizon model, let discounted price S be a martingale under the actual probability measure P and the desk's full information. For a predictable integrable strategy h with admissible gains, E[sum h_t (S_(t+1)-S_t)]=0. Deduct nonnegative trading costs and expected net gains are nonpositive. Faster computation, dynamic allocation, and adaptive indicators cannot overturn those assumptions. An equivalent martingale measure used in no-arbitrage pricing is different: it need not be P and does not imply zero physical expected risk premia. This example disproves a universal-profit promise; it does not assert that actual markets satisfy the example.

Use two objectives distinctly:

• **Policy objective:** maximize the desk's stated net portfolio utility subject to constraints. It can favor a hedge with negative standalone expected P&L.

• **Change criterion:** require a proposed intervention to earn its complexity, switching costs, and uncertainty relative to a frozen feasible incumbent.

Improve daily *opportunity coverage*: inspect more feasible, economically distinct opportunities before their deadlines. Do not target a daily trade quota, activity, or a guaranteed positive day. Include doing nothing, retaining a position, reducing it, and buying information as distinct decisions. Abstention from a new order is not liquidation.

2. State, beliefs, and five budgets

Maintain a decision snapshot with these components:

Component	Required contents
Information	As-of observations, source lineage, revisions, feature availability, quote age, missingness, uncertainty in clock offsets
Market belief	Conditional payoff/fill/cost distributions; latent-state probabilities; model alternatives; coverage gaps
Own state	Reconciled positions, cash, liabilities, pending orders, reservations, possible fills, collateral by venue
Policy state	Immutable policy/model/config IDs, approved actions, evidence stage, incumbents, elapsed holds and deadlines
Learning state	Matured outcomes, pending labels, trial ledger, estimands, test budgets, deployment history
Resources	Risk and funding headroom, executable capacity, compute budget, information acquisition budget, remaining opportunity time

Spend five separate budgets: capital/loss capacity, liquidity/market impact, time/compute, information acquisition, and statistical false-decision tolerance. Money is not interchangeable with statistical error probability. Express monetary comparisons in a common currency/horizon; keep hard limits separate from utility penalties.

Model a latent state z using a **filtered** belief b_t(z)=P(z_t=z | information actually available at t). In a specified state-space model, update via predictive transition followed by the new observation's likelihood. For example:

$$b_t(z) \propto p(y_t \mid z, a_{t-1}) \sum_{z'} p(z \mid z', a_{t-1}) b_{t-1}(z').$$

Include actions in transitions or observation models when they affect impact, inventory, fills, or future opportunity. The equation specifies a model; it does not make its likelihoods known. Check predictive calibration, sensitivity to state definitions, and a simpler nonregime baseline. Fit state definitions, transition parameters and every preprocessing step only on data then available; causal filtering with full-sample fitted parameters still leaks. Do not use future-smoothed states, retrospectively named regimes, or hard classification confidence as historical live knowledge [C02].

Prefer uncertainty over *action value* to uncertainty over narrative labels. Two plausible market explanations may recommend the same small trade; one confident regime label may still leave execution value unresolved. Distinguish:

• outcome randomness, which may be paid risk;

• parameter/model uncertainty, requiring sensitivity or a defensible ambiguity set;

• data/clock uncertainty, invalidating some inputs;

• own-state uncertainty, requiring reconciliation or set-valued risk checks;

• identification failure, where observed data cannot answer the counterfactual at all.

Do not compress these into one model-confidence number. A Bayesian credible interval, sampling confidence bound, stress scenario, and ambiguity set have different meanings.

3. Feasible actions and portfolio value

Construct a small approved menu: continue the incumbent; skip a new entry; change size within approved bounds; select execution mode; reduce or hedge existing risk; request a useful observation; or invoke a contingency. The menu must respect existing entry-clock authority. Candidate generation cannot silently authorize new instruments or live exploration.

For each candidate policy pi, model complete net portfolio wealth paths W(pi), including open-order outcomes, fees, spreads, impact, financing, collateral, holding/exit costs, and instrument semantics. Compare identical initial wealth, calendar, and external scenarios. Represent each material cash cost once. If a conditional fill payoff already includes adverse selection, do not subtract a second adverse-selection penalty.

An illustrative robust planning objective is:

$$J_t(\pi) = \inf_{P \in \mathcal{P}_t} E_P[U(W^{\pi}_{t+H}) \mid \mathcal{I}_t].$$

Choose U and the feasible path set from the mandate, with explicit interim capital, liquidity, concentration, turnover, and operational constraints. If using expected P&L minus a separate risk penalty, specify that as an alternative objective with consistent units. Do not add unexplained variance, entropy, Sharpe, confidence, and dollars. Include terminal liquidation or residual inventory value so a short planning horizon cannot hide costs beyond its endpoint. No modeled scenario set guarantees protection against all real-world losses [C01, C05].

For a **conservative improvement gate** against feasible incumbent pi0, evaluate the same-model paired difference:

$$G_t(\pi; \pi_0) = \inf_{P \in \mathcal{P}_t} E_P[U(W^{\pi}) - U(W^{\pi_0}) \mid \mathcal{I}_t].$$

Select an action only when G exceeds a declared hurdle and all evidence, timing, and feasibility gates pass. The hurdle covers a required economic margin or additional policy-change friction **not already charged** in W. It is not an arbitrary number claimed to be learned. Statistical uncertainty can define a justified set or a separate evidence gate; distinguish that from economically adverse outcomes.

Do not substitute min_P E[U(W^pi)] minus min_P E[U(W^pi0)]. Their worst cases can differ. **Checked illustration:** with two scenarios, candidate utilities (5,1), incumbent (0,4), and ambiguity allowing either scenario with certainty, separate minima give +1, while the paired robust difference is min(5,-3)=-3. The candidate improves worst-case *levels* but does not dominate the incumbent across models. Both objectives can be legitimate; name which was selected. The paired gate is deliberately conservative and may reject useful risk redistribution. It is not universal optimality.

Treat the entire desk jointly. Two acceptable sleeve additions may violate a common funding or tail limit when combined. Reserve resources atomically before submission. If proposing marginal allocations, recompute shared constraints after each accepted action, or solve a joint batch. Opposing uncertain orders cannot be assumed to cancel each other before fills.

Use capital shadow prices only when the underlying optimization supports them. For differentiable concave net value g_i(q_i) and shared linear resource limits A q <= b, an interior optimum satisfies marginal benefit g_i'(q_i)=(A^T lambda)_i. The multipliers price scarce resources **locally within that model**. With nonconvex lots, discontinuities, nonlinear margin, or uncertain liquidity, use finite perturbations and explicit feasible comparisons instead of false precision. Capacity changes the opportunity, not only position size [E04, E05].

4. Decision boundaries and the value of computation

Prioritize research where uncertainty could change the best feasible action before its deadline. High entropy about an irrelevant fact deserves no extra work; a narrow uncertainty interval around break-even may deserve immediate attention. This is a decision-theoretic principle, not permission to invent an information-value estimate.

For an optional computation or observation c taking time tau, estimate:

$$VOC(c) = E[V_{t+\tau}^{after\ c}] - V_t^{act\ now} - K_c.$$

Use the same terminal date and continuation rights: acting now may still allow learning later. Both values include attainable execution, market evolution during delay, and comparable portfolio utility. K_c must be in matching utility units; for nonlinear U, put monetary compute/acquisition costs inside wealth or justify the conversion instead of subtracting raw dollars. Do not charge delay loss again if it is already inside the future value. Evaluate against *acting now*, not only against waiting without computing. Estimate routing benefits from earlier cases, then test the complete routing policy prospectively [O04, C07].

Use an anytime pipeline: compute the cheapest admissible baseline; retain its feasible action; refine only while conservative estimated decision improvement exceeds cost and a worst-case completion time fits the deadline. Stop early when defensible action-value bounds establish the same choice across relevant uncertainty, or no affordable computation can change the action. Heuristic intervals are not mathematical certificates. A negative one-step VOC does not rule out valuable complementary computations; evaluate a small affordable bundle menu when warranted. Bound the controller's own analysis cost: recursively debating computation can consume the entire opportunity.

Hypothetical arithmetic: candidate A has scenario incremental utilities (8,-3), and probability of scenario 1 lies in [0.45,0.60]. Worst mean improvement is 1.95. If the uncharged change hurdle is 2, keep the incumbent. Candidate B has (4,1), giving worst mean 2.35; if feasible, it passes that illustrative gate. This favors robustness of the decision, not the most vivid forecast. No market measurements are implied.

Forecast scoring remains necessary even with decision-focused training. Training only against trading P&L can learn simulator quirks, omit untraded observations, or hide miscalibration. Compare proper forecast scores, decision regret on a defined feasible problem, and realized net-policy utility separately [E01].

5. Receding-horizon planning and restrained adaptation

Use three clocks:

1. **Execution clock:** validate data and own state, select among approved actions, reserve resources, submit and reconcile. Honor a hard deadline.

2. **Belief clock:** update forecasts, costs, fill models, and filtered state within a frozen, tested update rule. Trigger only on materially new information or prescribed reviews.

3. **Policy clock:** propose a new rule, state representation, model family, adaptation rate, or action universe; replay and validate it before promotion.

An approved policy may adapt every event. That does not authorize rewriting it after every loss. A rollback, learning window change, reset, or new ensemble weight rule is itself a consequential research choice. Keep adaptation within explicit bounded envelopes; changes outside them become new versions.

When inventory or costs couple decisions over time, plan over H steps but execute only the first action, then replan from reconciled state. Keep horizon, terminal inventory treatment, uncertainty model, and fallback fixed during evaluation. Scenario-tree decisions cannot use future branch identities before

those observations arrive. A deterministic open-loop plan is an approximation, not full contingent control [C01].

Avoid unnecessary churn. Let transaction costs, uncertainty, and existing positions create a no-trade region. For the one-asset objective F(x+delta)-k*abs(delta) with differentiable concave F and no other constraints, zero change is optimal when abs(F'(x))<=k; constraints alter the condition. Quadratic costs often imply gradual adjustment rather than an exact no-trade band. A cooldown or separate entry/exit hurdle is a policy choice to test, not a substitute for the actual cost model [C03; earlier S3].

Use robust dynamic programming only with its required ambiguity assumptions. In particular, stagewise adversarial choices can create impossible combinations when uncertainty is coupled across time. Rectangularity supports specific recursive formulations; otherwise retain coherent path/model uncertainty and avoid claiming a Bellman optimality guarantee [C06].

Recognize dual control: an action can earn payoff and reveal information. Passive paper observation is preferable when it answers the question, but it cannot reveal actual queue fills or market impact. Any live experiment must be separately within user authorization and mandate, have bounded capital and a prespecified design, and account for future-state effects. No information bonus overrides hard risk limits. Approximate dual-control theory is not an unrestricted reinforcement-learning license [C04].

6. Master procedure

Apply the following procedure to a desk, strategy, or sleeve. Unknown quantities become explicit research dependencies; do not hallucinate inputs to complete the algorithm.

```
CONFIGURE once per approved version:
  freeze economic contracts, incumbent, objective and benchmark
  define instruments, action menu, timing authority and budgets
  define as-of data rules, update rules, scenarios and hard limits
  define evidence gates, deadlines, fallback and contingency policies
  define learning estimands, label maturity and error spending

ON an eligible information, market, timer, or own-state event:
  1 ingest/deduplicate; retain publication and receipt history
  2 reconcile own state; update all possible open-order exposures
  3 if hard integrity/risk conditions fail:
      block alpha expansion; invoke approved specific contingency
      persist cause; RETURN from alpha path or enter risk-only mode
      do not infer flat positions or blind-liquidate
  4 update filtered beliefs using only presently available evidence
  5 build a small feasible, approved menu including the incumbent
  6 compute cheap fill-aware, portfolio-aware net action values
  7 remove unsupported, stale, expired or infeasible candidates
  8 refine only where estimated value of computation justifies delay
  9 compare paired improvement, uncertainty, hurdle and constraints
 10 if a candidate passes, atomically revalidate current dependencies,
    policy, own state and limits; reserve resources and log intent
    on version conflict recompute; never commit the stale decision
    else retain feasible incumbent and record why no change passed
 11 execute through deterministic lifecycle; recheck expiry and all
    required pre-send validity conditions at the submission boundary
 12 reconcile fills/cancels; update reservations from actual events
 13 append decision, alternatives, information and policy versions

ON an outcome-maturity or learning event:
  14 score forecasts and all eligible decisions; track censored labels
  15 update prespecified evidence monitors with valid information set
  16 attribute changes to forecasts, capture, costs, risk or mechanism
  17 route to continue/reduce/pause/investigate/retire per approved rules
  18 create a new challenger only for an identified, testable bottleneck
  19 evaluate the entire adaptive policy against frozen alternatives
  20 promote only to the next authorized evidence stage; retain rollback
```

If the incumbent is infeasible, the alpha comparison is no longer the decision. Invoke the appropriate risk/operational contingency. Risk reduction may rationally have negative expected profit. Maintain separate emergency authority, available reliable data, and execution limits.

7. Research opportunities and falsification

These are proposed combinations, not verified new discoveries or proven trades:

Hypothesis	Minimal discriminating experiment	Failure criterion
Spend analysis on decision-boundary cases	Compare cheap, always-deep, and budgeted refinement on identical as-of events	Added utility fails to cover compute and latency
Separate information decay from capture decay	Hold forecast fixed; compare execution modes and delayed entries	Improvement vanishes with attainable fills and costs
Allocate by shared capital scarcity	Replay joint portfolio with endogenous reservations versus independent sleeve caps	Gain disappears under common stress and collateral constraints
Preserve liquidity for later opportunities	Compare full deployment with a trained reserve rule on the actual future opportunity stream	Gain depends on future knowledge or assumed unlimited reinvestment
Adapt to observable mechanisms, not outcome labels	Compare frozen, calendar-refit, and evidence-triggered policies	Detection delay, false alarms and switching erase benefit
Use simpler actions when uncertainty is irrelevant	Compare action-gap routing with entropy-only routing	No budget-matched benefit or fragile threshold dependence

Waiting has option value when commitment is costly or irreversible and useful learning/opportunities may arrive. It also has decay and missed-fill cost; a positive waiting value in one investment model is not a universal reason to delay trades [E06].

Test transfer across environments only after defining environments independently of favorable outcomes. Stable relations can still be confounded; causal invariance results require structural assumptions, including restrictions on interventions affecting the target [X01]. Treat invariance tests as falsification tools unless identification assumptions are justified. Competition can change the target relation itself.

Trading can alter its own training distribution through impact, information leakage, inventory and opponent reaction. Distinguish external market drift from policy-induced change; log policy versions and sizes as state. Stable repeated retraining is not necessarily the best deployed policy, and may converge poorly or fail under violated assumptions [E02, E03].

8. Minimal implementation and measurement

Start with one event log, one deterministic state reducer, one small candidate scorer, one portfolio-risk/reservation authority, one order manager, and one replay/evaluation worker. Use a modular process before introducing distributed services. Cache shared features with version and availability keys; invalidate dependencies when their facts change. Add richer latent-state models, solvers, or parallelism only after a measured bottleneck justifies them.

The bundled `scripts/adaptive_arbiter.py` is an offline finite-menu reference kernel for the paired robust gate and scenario constraints. It consumes supplied utilities, beliefs and load scenarios; it neither estimates them nor submits orders. Its tests validate specified control invariants using synthetic inputs. Read its module contract before using it. It is not the whole master procedure, a certified risk engine, or empirical proof of edge.

Measure improvement on a declared baseline:

Requested quality	Operational measurement
Intelligence and accuracy	Proper forecast scores, decision errors near thresholds, net portfolio utility with uncertainty
Awareness	Data/clock lineage coverage, unresolved-state age, reconciliation mismatches detected before action
Responsiveness	Event-to-valid-decision latency and change-detection delay at a fixed false-alarm budget
Speed and bandwidth	Tail latency, deadline misses, feasible opportunities evaluated per unit compute, capacity after cost
Precision	Calibration in action-relevant bins, fill-conditioned markouts, reservation and settlement correctness
Dynamism	Held-out utility of the entire adaptive policy after turnover, false alarms, retraining and switching
Efficiency	Net value per bounded resource plus total value; avoid maximizing ratios that sacrifice useful scale
Edge	After-cost expected advantage over the stated benchmark and constraints, with evidence stage

Report percentage or multiple improvements only for measured metrics with numerator, denominator, workload and uncertainty. A faster synthetic function is not a more profitable strategy. A broad research procedure can improve the quality of any assessment; some assessments should end with repair, replacement, or no trade.

3 / Learning and evidence

1. Evidence contracts and feedback types

Before learning, specify **what quantity is identifiable** from the available observations. A strategy may forecast a common outcome for every candidate, yet reveal executions only for submitted orders. One-step routing rewards and full inventory trajectories are different targets.

Every evidence object must carry:

```
policy/version and candidate family
estimand, units, economic hurdle, null and alternative
eligible population, opportunity-generation rule and benchmark
decision, publication, receipt, observation and outcome-maturity times
information available to detector AND to stopping/routing controller
conditional dependence/tail assumptions and support requirements
estimator, nuisance fitting, tuning and missingness rules
false-decision budget, repeated looks, resets and multiplicity plan
maximum sample/horizon if applicable; detectable economic change
permitted consequence: investigate / reduce / promote / retire
```

Reject confidence claims without this contract. Do not combine a Bayesian posterior, an ambiguity-set minimum, a fixed-time confidence interval and a stress percentile into one confidence score. A drift detector is an alert for a specified change, not a diagnosis of its economic cause.

Keep production, challenger and evidence registries distinct. Production includes a frozen adaptive update rule and its tested ranges. Challenger versions are immutable. Each evidence contract belongs to a specific estimand/version and records every outcome inspected. Changing a reward, target policy, learning rate, reset rule or horizon is a research choice that may require a new evaluation.

2. Counterfactual support

Distinguish these feedback regimes:

Data available	Permitted comparison	Missing counterfactual
Common resolved event with negligible own influence	Preissued forecasts against the same outcome	Alternative execution price and fill
Supported randomized route with known selection probabilities	One-step policy reward under identification assumptions	Unsupported actions; changed future inventory state
Quote/order-book replay	Policy performance in the specified simulator	True queue priority, impact, response to unplaced orders
Stateful portfolio actions	Trajectory value under an identified transition model or suitable experiment	Alternative trajectories from one-step weights alone

For discrete actions, let $b_t(a|x)$ be the actual logging probability and $\pi(a|x)$ the target probability. Require $\pi(a|x)>0$ only where $b_t(a|x)>0$ for relevant contexts, plus the causal/consistency assumptions of the evaluation. Known propensities do not fix unobserved confounding or action-induced future-state change [L03, L08].

A standard doubly robust pseudo-outcome is:

$$\psi_t(\pi) = \sum_a \pi(a \mid x_t)\widehat{m}_t(x_t, a) + \frac{\pi(a_t \mid x_t)}{b_t(a_t \mid x_t)}[r_t - \widehat{m}_t(x_t, a_t)].$$

Fit reward models on earlier data or use a dependence-valid honest evaluation design. The robustness concerns particular nuisance-model errors under the model; it cannot recover an action with zero support. Examine overlap, extreme weights and sensitivity. Weight clipping changes bias; an effective-sample-size diagnostic is not a validity certificate. Do not reconstruct propensities from observed frequencies when the actual assignment mechanism was deterministic or different [L04].

Log context, feasible menu, action, actual propensity, randomization ID where applicable, own-state snapshot, policy version, reward definition, maturity, costs, and observed/simulated provenance. Preserve rejected opportunities, including no trades and nonfills. A deterministic log choosing passive only in quiet markets and aggressive only in volatile markets does not identify their comparison in either market through importance weighting. A model can extrapolate, but must report its additional assumptions.

Specify three portfolio counterfactuals separately:

• Freeze other sleeves and add/change the candidate: marginal contribution under fixed peers.

• Reoptimize peers under the same shared limits: replacement value plus reallocation effects.

• Change execution/size enough to alter prices or future state: deployment-dependent value requiring impact/transition assumptions.

Do not sum standalone sleeve improvements when nonlinear risk, competition for capital, or impact makes them nonadditive. Common shocks used for paired simulation improve comparability only when the action-dependent simulator defines both paths coherently.

Choose the least costly identifiable experiment. Shadow observation is useful for data, timing and forecast checks; it is not proof of hypothetical fills. A randomized real-money experiment requires explicit authorization and bounded risk/exposure, design, assignment and abort rules. Research permission alone does not supply it.

3. Sequential evidence and its information clock

Fixed-time tests generally cannot be repeatedly checked at nominal significance without an appropriate sequential or multiplicity design. Confidence sequences and e-processes provide tools under explicit conditional assumptions [L01, L02]. Record the target: constant mean, cumulative average conditional value, recent-window mean, or change since an epoch. A lower bound for average value since inception does **not** establish present or future edge after deterioration.

Elementary derivation, deliberately limited. Let resolved increments satisfy -1<=X_t<=1 and test the strong null E[X_t|F_(t-1)]<=0 for every t, where F is the full information used to select observations and stopping decisions. Choose lambda_t in [0,1) measurably before X_t. Define:

$$E_0 = 1, \qquad E_t = \prod_{i=1}^{t}(1 + \lambda_i X_i).$$

The factors are nonnegative and conditional expectation cannot increase E_t under the null. Thus it is a nonnegative test supermartingale, and the chance it ever reaches 1/alpha under that null is at most alpha. Use log1p for stable computation. This derivation is an illustrative specialization, not the recommended universal trading test.

Crossing rejects the claim that conditional improvement was never positive over the tested sequence. It does not test an arbitrary average-mean null, prove the current mean is positive, or produce a posterior probability of future profit. For an intended cumulative-average or time-varying target, use a construction valid for that target [L03]. If D_t is a net improvement and h a hurdle, X_t=(D_t-h)/B requires a defensible bound abs(D_t-h)<=B; a historical maximum or stop order does not establish it. Clipping creates a different estimand and can conceal tail loss. For unbounded P&L, use justified conditional-tail assumptions or a method designed for them; do not fake boundedness.

Information timing is part of validity. An e-process justified when the observer sees a coarse history may be invalid when a richer-history controller decides when to stop. Delayed multi-horizon forecasts, partial outcomes and intraday prices can enlarge the actual information set. Validity must hold in that full decision filtration, or a justified adjustment is required. Fixed weighted averages can combine valid e-processes for the same null in the same filtration; arbitrary products, hindsight-selected weights, and different filtrations need additional justification [L09].

Allocate error across monitors and epochs prospectively. For detector weights w_j>=0 with sum_j w_j<=1 and restart epoch r=1,2,..., one transparent allocation is alpha_(j,r)=alpha_total*w_j/[r(r+1)]. The total is at most alpha_total, so a union bound controls the family of individually valid false rejections without requiring independence. This is conservative accounting, not an optimal allocation. It does not repair invalid individual tests or retrospective epoch selection. A reset after failure does not erase the failed history.

The best among many ordinary pointwise lower bounds is not automatically a valid bound for the selected candidate. Use simultaneous/selection-aware inference or independent confirmation. Anytime validity for one stream does not eliminate strategy-search bias, multiple detectors, or model misspecification.

4. Drift, delayed feedback, and retirement

Attach outcomes to the original decision version. Maintain pending labels with planned horizon, resolution rule, censoring reason, and informative-delay flags. Never treat unresolved outcomes as zero or score only quick winners. A position's closing time and its evaluation horizon may differ. Compare mature cohorts under the frozen definition, including appropriate mark-to-market outcomes and separately accounted pending risk. Delayed-feedback results require their own assumptions; trade duration is often action- and outcome-dependent [L07].

Overlapping outcomes are not independent updates. Numbering trades by settlement time does not establish a conditional-mean null. Use a defensible delayed-observation method or prespecified evaluation blocks with suitable dependence/tail assumptions. Retain partial information available to the controller in the evidence contract. Fewer, honestly identified opportunities can be more informative than thousands of overlapping timestamps.

Route alerts by the layer that failed:

Signal	First diagnostic	Candidate response
Missing sequence, stale quote, unknown inventory	Operational integrity	Disable affected expansion; reconcile immediately
Inputs change but conditional forecast scores hold	Opportunity mix/support	Review coverage and supported reweighting
Mature comparable forecasts deteriorate	Forecast/model stability	Examine calibration, feature lineage and challenger
Forecast stable, filled markouts worsen	Execution/capture	Inspect queue, latency, spread, size and selection
Gross payoff stable, shared stress rises	Compensation/portfolio suitability	Reprice capital, scenario risk and size
Completed outcomes improve as pending losses grow	Maturity selection	Repair evaluation before attributing edge
Size/routing changes precede changed outcomes	Policy feedback	Compare deployment cohorts and impact hypotheses

Operational intervention need not wait for statistical significance. An evidence alert usually calls for investigation or a prespecified reduction, not an automatic new strategy. The counterfactual cost of false alarms, delayed detection, missing opportunities, and switching belongs in the whole-policy evaluation.

Forgetting has a cost. Short windows adapt quickly but discard evidence; long windows gain precision but react slowly. Nonstationary-bandit theory makes this dependence on variation explicit, under its setting. Regret against the best available arm is relative: all arms can lose money, and arbitrary rapid change can defeat tracking [L06]. Preserve cash and a feasible incumbent benchmark. A conservative-bandit bound on cumulative expected reward is not a realized-loss, margin, or drawdown guarantee. Signed trading rewards also do not inherit a nonnegative-reward multiplicative baseline constraint unchanged [L05].

Treat recent-window length, reset threshold, ensemble weights and learning rates as parameters of the evaluated policy. No universal setting is best for every market. Prefer a small preregistered family and mechanism-grounded shifts to a large daily winner search. Evidence of no useful benefit can justify retirement; insufficient precision is a separate conclusion.

5. Experiments and acceptance gates

Compare at least the frozen baseline, a simple periodic-refit baseline, and the proposed adaptive policy, replaying every forecast, update, reset, resource reservation and switch chronologically. Charge data, computation, latency, missed fills, turnover, funding and maintenance where relevant. Preserve the original opportunity population and common evaluation horizon.

Test these failure cases before trusting an adaptive claim:

- Zero-propensity target actions, simulator-only fills, or an unlogged action menu.

- Strong historical average evidence followed by a change in present economics.
- Overlapping delayed labels, quick-winner completion and outcome-dependent censoring.
- Multiple monitors, resets and holdout reuse without declared error/search accounting.
- Full-sample fitted regimes despite apparently causal filtering.
- Low regret against losing arms, or positive gross returns erased by switching.
- Distinct sleeves failing together through one liquidity or collateral dependency.

Promote only when candidate-specific economic evidence, realistic execution, portfolio constraints and operational replay pass their declared gates. If the data cannot identify the desired improvement, name the missing counterfactual and the least expensive experiment that can. An honest inability to identify an advantage is useful information; a more elaborate model is not automatically the remedy.

4 / Runtime and execution

1. Minimal architecture and decision contract

Start with a modular service containing an append-only event journal, deterministic state reducer, incremental features/forecasts, finite action scorer, portfolio reservation authority and order-lifecycle manager. Run outcome evaluation and research asynchronously. The economic policy and safety-critical execution path should not require a synchronous LLM response. Add process boundaries only for demonstrated isolation, throughput or reliability needs [O05, O06].

Every decision needs a reproducible contract:

```
decision_id, opportunity_id, policy/model/config hashes
as_of_snapshot_id, source/revision/dependency IDs
event/publication/receipt/feature-ready times and clock-error bounds
action-specific required inputs, freshness rules, evidence stage
reconciled positions, cash, pending orders, possible-fill envelope
candidate menu, incumbent continuation, scores, units, rejected reasons
hard limits, joint portfolio reservation version, deadline
intent_id, payload_hash, reservation_id, venue deduplication contract
fallback and recovery policy, expiry/revalidation rules
realized fills/costs/outcomes, corrections, attribution and maturity
```

Produce the cheap feasible baseline before optional refinement. No mandatory risk, contract-semantic or data-integrity check becomes optional merely because its expected dollar value is difficult to estimate. A fast invalid decision is not throughput.

2. Data and clock semantics

A feature is usable only after all required source facts have arrived and computation has completed. Event time is not information availability. Preserve publication time, receipt time, revision ID, feed sequence and feature-ready time. Record what the strategy actually saw separately from later corrected truth.

Specify provisional versus finalized bars/windows. Respect a designated closed-bar clock. A watermark represents stream progress under assumptions; it does not prove every relevant observation has arrived. Late data can update future decisions through a declared correction policy, but cannot rewrite a prior decision snapshot [O01]. Missing values, true zeros, stale values and failed extraction are distinct states.

Cache shared features by dependency/revision IDs, code/config/model hashes and relevant cutoff. Invalidate descendants of changed facts, while leaving unrelated calculations alone. Revalidate required freshness and own state at the submission boundary even after a cache hit. Keep terminal/settled contract definitions versioned.

Store physical timestamp uncertainty and causal sequence separately. A deterministic total order is useful for replay, but cannot establish true cross-feed lead-lag or physical causality. Do not claim a 2 ms information lead when clock uncertainty is 5 ms. Prefer server sequence and acknowledgement dependencies where appropriate; the actual clock-quality and venue contract determine interpretation [O07].

3. Orders, uncertain exposure and reservations

Represent possible account states when an order outcome is uncertain. Use reconciled positions plus feasible outstanding-fill paths, preserving dependence within order chains. An unacknowledged accepted order can already be exposed. A cancellation request is not a cancellation; a timeout is not a rejection [O02].

For expansion candidate a, check every modeled feasible own-state x and execution path f:

`constraints(next_state(x,a,f)) must hold`.

Use a conservative tractable envelope if exact enumeration is too expensive, document its looseness, and maintain a reconciliation path. This is protection against stated possibilities, not unknown venue behavior. If the incumbent already violates a limit, invoke the separate approved recovery objective; requiring all recovery states to be instantly compliant can deadlock necessary reduction.

Hypothetical example: certainly long 40 units, with an additional 60 possibly filled. An 80-unit sell intended to flatten can create a 40-unit short. Under simplified long-only assumptions, selling 40 would leave 0-60 long, but even that requires checking other working orders and current venue rules. Reconcile first; do not assume the midpoint of the uncertainty interval is the position.

Maintain lifecycle, cumulative fill quantity, remaining executable quantity, correction state and reconciliation confidence independently. A high-precedence status label can hide relevant partial fills. Release reservations only after terminal facts and quantities reconcile. Handle execution corrections/busts as explicit events.

For a 100-unit order with 40 filled, a pending reduction to total 70 can still allow the old remaining 60 until accepted. A pending increase to total 150 can allow up to 110 additional units if accepted. An atomic amendment follows one chain; it is not automatically two independent full orders. A separately submitted replacement before cancellation confirmation can expose both. Generate the feasible set from verified protocol semantics, not a generic netting assumption.

Serialize shared capital and risk reservations or use an equivalent atomic coordinator. Two sleeves cannot each consume the same available headroom. Bind intent identity, payload hash, policy/snapshot version and reservation in a durable transaction before sending. Immediately revalidate current dependency versions, limits and expiry. On conflict, recompute; do not commit a stale optimization result.

Use a stable client intent ID for retries only when the venue enforces compatible idempotency and retention semantics. Reusing an ID with changed arguments may be invalid; a new ID may create another order. If submission outcome is ambiguous and deduplication cannot be proved, lookup/reconcile rather than blindly resend. Local exactly-once processing is not exactly-once external execution [O03].

4. Capability-specific degradation

Build an action-to-dependency map. Partial failure need not disable unrelated capabilities, but any fallback must already be an evaluated policy with valid inputs.

Failure	Response under the approved contract
Optional language model is late	Use the approved simpler policy if still valid; never silently impute its missing feature

Failure	Response under the approved contract
Required entry feed is stale	Disable affected new entries; keep valid reconciliation/protection routes
Position or order state is unknown	Retain reservations, query, reconcile; use only explicitly justified recovery
Optimizer misses deadline	Revalidate the feasible fallback or abstain from expansion; never send an expired target
Current limit is breached	Run the specific risk-reduction contingency; ordinary hysteresis cannot delay it
Statistical evidence weakens	Follow the declared review rule; do not rewrite live policy in response to a story

Track distinct states: READY, LIMITED_CAPABILITY, RECONCILING, RISK_RECOVERY, and DISABLED_FOR_EXPANSION. Define entry and recovery conditions for each. These states are not interchangeable with portfolio positions. A state disabling new entries can coexist with active position management. Restoring data connectivity does not by itself prove account reconciliation or authorize resuming risk.

5. Latency and resource efficiency

Prioritize execution events and risk reconciliation over optional research. Bound queue lengths and define backpressure. Coalesce redundant market updates only when strategy semantics permit; never silently discard fills, cancellations or corrections. Maintain maximum observation age as well as processing speed.

Measure publication-to-receipt, receipt-to-feature, feature-to-decision, decision-to-reservation, submit-to-ack and ack-to-fill, with clock uncertainty. Report end-to-end p50/p95/p99, deadline misses, queue age, and executable economic capture at size. Do not sum component percentiles and call the result the system percentile. Fan-out and shared-resource contention can make tail performance worse despite a faster average model [O05].

Give optional computation a deadline budget derived from opportunity decay, plus time for mandatory validation and submission. Never use hedged duplicate **mutating order** requests as a casual transfer of redundant read-request techniques. Restrict parallel work to independent read/research tasks or properly coordinated state changes.

Remove unused features, models and branches when held-out incremental value fails to cover latency, data licensing, testing, monitoring and maintenance. A small shared feature store with clear lineage can outperform independent per-agent recomputation. Avoid multiplying identical evidence through redundant models. Track who consumes each input and which policies depend on it [O06].

6. Replay and failure acceptance

Replay must reproduce decisions using the original available data, policy versions, update schedule, tie rules and numeric tolerances. Corrections produce new events; they do not mutate the past. Matching replay is an operational requirement, not evidence of profitable prediction.

Inject at least these failures in a sandbox adapter before live integration:

1. An early exchange timestamp arriving after the decision cutoff.

2. A revised input with an old dependent cache entry.

3. Acceptance followed by a crash before acknowledgement.

4. Fill after cancel request; cancel rejection after complete execution.

5. Upward/downward amend, crossed replies, and separate cancel/new order.

6. Concurrent sleeves competing for the same collateral/reservation.

7. Unknown [40,100] position and an unsafe 80-unit flatten instruction.

8. Optional model outage with valid protection and account feeds.

9. Optimizer delay, queue overload and a fallback that has itself expired.

10. Execution correction/bust after apparent terminal completion.

Acceptance means expected state transitions, conserved quantity/cash accounting under the venue model, no duplicate semantic intent, and no unauthorized expansion. Test the real adapter protocol; a pure selector's unit tests cannot establish these distributed-system properties. Reconcile after restart before claiming restored readiness. Measure any speed improvement on representative workloads and compare economic outcomes at matched capital, not only microbenchmarks.

Offline kernel contract and use

Run the synthetic invariant tests with `python -m unittest discover -s scripts -p test_adaptive_arbiter.py -v` from the skill directory. Import `select` and frozen dataclasses from `scripts/adaptive_arbiter.py`; call `select(policy, incumbent, candidates, now=..., decision_deadline=...)`. Inputs use integer UTC epoch seconds and named utility/constraint units. Test fixtures show minimal synthetic construction.

Bind policy/action IDs externally to immutable definitions and authenticated input snapshots. The module cannot authenticate a supplied evidence flag or prove that utilities correspond to an action. Utilities must account for all execution branches consistently with the supplied portfolio scenarios. Constraints use additive loads and conservative Cartesian combinations of outstanding-order alternatives; this can overreserve mutually exclusive order chains. Use a validated joint-state evaluator for nonlinear margin or exact chain semantics.

The pure function checks deadlines against supplied time, without enforcing wall-clock completion or atomic external reservation. Its combination cap bounds one dimension of work, not total runtime. Malformed inputs raise ValueError: the caller must fail closed to alpha expansion and continue the separate recovery path. Repeated calls yield the same proposal; downstream order deduplication, persistence, synchronization and reconciliation remain unimplemented. These are deliberate prototype limits, not completed production capabilities.

5 / Annotated primary sources

Research checked October 24, 2026 UTC. This pass adds **30 distinct primary-source entries** to the earlier 50-entry register. C08 is an explicit alias of earlier S3 and is not new. Sources include papers, author manuscripts, one official protocol specification, and official engineering articles. Alternate versions/URLs are not independent confirmations. This was targeted research and substantive inspection of specified sections, not an exhaustive literature review or proof-by-proof replication.

Controller design, examples, resource contracts and proposed experiments are original engineering synthesis or elementary derivations. A source's theory or historical result does not validate a proposed trading implementation. Where only abstracts were accessible, the entry says so. Verify current venue capabilities separately from historical specifications.

Control and computation

C01 - Boyd, Busseti, Diamond, Kahn, Koh, Nystrup and Speth

Multi-Period Trading via Convex Optimization. Foundations and Trends in Optimization 3(1), volume dated 2016, manuscript copyright 2017. Author PDF. DOI 10.1561/2400000023.

Inspected: abstract and sections on omitted features, forecast error, multi-period use and computation. **Supports:** constrained planning with costs/risk, followed by first-action execution and replanning. **Limit:** explicitly does not produce the forecasts; basic trading assumptions require extension for partial fills, queueing and latency. Convexity is a property of a particular formulation, not every desk problem.

C02 - Hamilton (2005)

Regime-Switching Models, author manuscript May 18, 2005 for Palgrave. Author PDF.

Inspected: model, filtering equations, smoothing and identification discussion. **Supports:** probabilities over unobserved states conditional on current information differ from retrospective smoothed probabilities. **Limit:** states and transition dynamics are model assumptions. Full-sample fitted parameters still leak future information even when the filtering recursion looks causal. Sparse transitions weaken estimates.

C03 - Davis and Norman (1990)

Portfolio Selection with Transaction Costs. Mathematics of Operations Research 15(4), 676-713. Publisher.

Inspected: publisher abstract/metadata; full-paper retrieval unsuccessful. **Supports:** a no-transaction region in the paper's consumption/investment model with proportional costs. **Limit:** do not import its boundaries into jumps, discrete contracts or arbitrary portfolios. The simple one-period derivative condition in the addendum is a separate elementary derivation.

C04 - Klenske and Hennig (2016)

Dual Control for Approximate Bayesian Reinforcement Learning. JMLR 17(127), 1-30. Primary PDF.

Inspected: introduction, controller construction, experiments 6.2-6.4 and conclusion. **Supports:** actions can influence payoff and future information; relevance and timing of learning matter. **Limit:** approximate control of simulated dynamical systems is not demonstrated financial alpha. Exact Bayesian control is difficult; an uncertainty bonus alone is not its solution or an authorization to experiment live.

C05 - Hansen and Sargent (2001)

Robust Control and Model Uncertainty. American Economic Review 91(2), 60-66. Author PDF.

Inspected: introduction, relative-entropy model uncertainty and time-consistency discussion. **Supports:** deliberate model-uncertainty preferences and constrained distortions around an approximating model. **Limit:** an ambiguity radius is not a posterior error probability. Resetting a whole-horizon distortion budget at every step can change the objective. Robustness only addresses the specified alternative models.

C06 - Iyengar (2005)

Robust Dynamic Programming. Mathematics of Operations Research 30(2), 257-280. Publisher.

Inspected: publisher abstract and author bibliography; not full proofs. **Supports:** rectangular uncertainty structure enables the analyzed robust dynamic-programming counterpart. **Limit:** a finite scenario scorer or arbitrary persistent-model uncertainty does not inherit those guarantees. Independently selecting each period's worst transition can construct incoherent paths.

C07 - Hay, Russell, Tolpin and Shimony (2012)

Selecting Computations: Theory and Applications. UAI. Author PDF; proceedings record.

Inspected: metalevel formulation, stopping results and assumptions. **Supports:** allocate computation by expected decision improvement; myopic evaluation can miss complementary sequences. **Limit:** specific bounds assume their stated utility/sampling model. Markets evolve during computation; no universal trading compute budget follows.

C08 - Alias of earlier S3

Gârleanu and Pedersen, *Dynamic Trading with Predictable Returns and Transaction Costs* (2013; inspected March 2012 author version). Author PDF.

Re-inspected: model and partial-adjustment proposition. Predictable signals with quadratic costs motivate gradual trading toward a forward-looking target. Proportional-cost no-trade regions are a different result. This is one existing source, not an additional confirmation.

Adaptive inference and learning

L01 - Howard, Ramdas, McAuliffe and Sekhon (2021)

Time-uniform, nonparametric, nonasymptotic confidence sequences. Annals of Statistics; inspected author revision August 2022. Primary PDF.

Inspected: conditional concentration construction, introduction and discussion. **Supports:** simultaneous coverage over time under specified conditional tail assumptions. **Limit:** nonparametric does not mean assumption-free; arbitrary heavy-tailed P&L is not bounded. Running intersections for a constant parameter cannot be blindly reused for changing targets. Coverage is not a promise about future profit.

L02 - Ramdas, Ruf, Larsson and Koolen (2022)

Admissible anytime-valid sequential inference must rely on nonnegative martingales. Inspected November 2022 author revision. Primary PDF.

Inspected: definitions and martingale/e-process framework. **Supports:** foundations connecting admissible anytime-valid procedures with nonnegative martingale methods. **Limit:** validity needs the null and actual information history. E-values are not posterior probabilities. The addendum's bounded-product example is an elementary specialization, with a deliberately strong conditional null.

L03 - Waudby-Smith, Wu, Ramdas, Karampatziakis and Mineiro (2024)

Anytime-valid off-policy inference for contextual bandits. Primary PDF, August 2024 revision; DOI.

Inspected: data assumptions, support, predictable tuning and time-varying values. **Supports:** adaptive logging and changing/dependent contexts under stated conditions; time-varying average conditional policy-value inference. **Limit:** target support and bounded reward requirements matter. Average value so far is not instantaneous edge. Endogenous future context from inventory or impact changes the interpretation of one-step comparisons.

L04 - Dudík, Langford and Li (2011)

Doubly Robust Policy Evaluation and Learning. ICML. Primary PDF.

Inspected: estimator, bias result and variance discussion. **Supports:** combining propensity weighting with reward prediction can reduce bias under the specified nuisance-model conditions. **Limit:** zero support, confounding, leakage or wrong payoff definitions are not repaired. Large weights may dominate uncertainty. Keep nuisance fitting and candidate selection honest under dependence.

L05 - Kazerouni, Ghavamzadeh, Abbasi-Yadkori and Van Roy (2017)

Conservative Contextual Linear Bandits. NeurIPS. Author PDF.

Inspected: assumptions, performance constraint, algorithm and theorem conditions. **Supports:** exploration with a modeled cumulative expected-reward baseline constraint under linear, bounded and conditional-noise assumptions. **Limit:** this is not a realized drawdown or collateral guarantee. Multiplicative constraints for nonnegative rewards cannot be copied directly into signed trading P&L.

L06 - Besbes, Gur and Zeevi (2014)

Stochastic Multi-Armed-Bandit Problem with Non-stationary Rewards. NeurIPS. Primary PDF.

Inspected: variation budget, regret benchmark, lower bound and restart policy. **Supports:** achievable tracking depends on reward variation and the balance between memory and forgetting. **Limit:** comparator regret is not absolute profit. Fast-enough changes can preclude sublinear dynamic regret in the model. Financial feedback and costs need separate treatment.

L07 - Joulani, György and Szepesvári (2013)

Online Learning under Delayed Feedback. ICML. Primary PDF.

Inspected: feedback protocol, delay assumptions and stochastic queue approach. **Supports:** feedback delays change learning guarantees. **Limit:** favorable delay conclusions need their stated assumptions. Trade duration may depend on action and payoff; quick-winner/slow-loser completion creates selection that cannot be ignored.

L08 - Li, Chu, Langford and Wang (2011)

Unbiased Offline Evaluation of Contextual-bandit-based News Article Recommendation Algorithms. WSDM; inspected March 2012 author revision. Primary PDF.

Inspected: replay algorithm and theorem assumptions. **Supports:** matched-action replay under randomized logging and the stated i.i.d. setting. **Limit:** market-price replay is a simulator, not equivalent randomized logged executions; absent queue positions and impact remain unobserved. More replay cannot supply missing support.

L09 - Choe and Ramdas (2026 accepted manuscript)

Combining Evidence Across Filtrations. March 15, 2026 revision of 2024 preprint; accepted JRSS B per author record. Primary PDF; version record.

Inspected: definitions, financial example, combination and lifting results. **Supports:** coarse-history validity need not survive stopping using richer information; justified adjustment can lift processes across filtrations. **Limit:** arbitrary products or combinations across incompatible histories are invalid. Common-null, common-filtration averaging has different conditions from such combinations.

Economics and feedback

E01 - Elmachtoub and Grigas (2022)

Smart "Predict, then Optimize". Management Science 68(1), 9-26; inspected November 19, 2020 arXiv v5. Primary full text; DOI.

Inspected: losses, framework and consistency assumptions. **Supports:** train predictions using downstream decision loss in a specified optimization problem. **Limit:** the consistency argument has continuity, symmetry and feasible-set conditions; unknown impact or drift is not automatically covered. A decision score is not necessarily a calibrated probability. Keep independent forecast and policy evaluation.

E02 - Perdomo, Zrnic, Mendler-Dünner and Hardt (2020)

Performative Prediction. PMLR 119. Primary PDF.

Inspected: definitions, convergence and assumptions. **Supports:** deployment can change the distribution being predicted. **Limit:** convergence requires model regularity and sufficiently weak distributional sensitivity in the analyzed result. Stability and optimality differ; an externally changing market need not follow a fixed deployment-to-distribution map.

E03 - Miller, Perdomo and Zrnic (2021)

Outside the Echo Chamber: Optimizing the Performative Risk. PMLR 139. Primary PDF.

Inspected: constructive counterexample, assumptions and optimization discussion. **Supports:** stable retraining can be very suboptimal for deployment-dependent outcomes. **Limit:** improving induced outcomes requires more structure than ordinary loss minimization. The result motivates evaluation against deployed consequences, not unrestricted live exploration.

E04 - Berk and Green (2004)

Mutual Fund Flows and Performance in Rational Markets. Inspected author manuscript October 15, 2002. Primary manuscript; author institution.

Inspected: opening argument and increasing-cost/decreasing-return model. **Supports:** ability can coexist with limited persistent investor abnormal return as capital competes and scale changes efficiency. **Limit:** delegated-management equilibrium assumptions are not an estimated desk capacity curve. Distinguish captured dollars, return per dollar, fees and scarce-capital surplus.

E05 - Greenwood and Thesmar (2011)

Stock Price Fragility. Journal of Financial Economics 102(3), 471-490. Article PDF; DOI.

Inspected: fragility/co-fragility construction, sample and endogeneity cautions. **Supports:** common ownership and correlated liquidity needs can relate to volatility/comovement. **Limit:** historical US 1990-2007 ownership evidence is incomplete and lagged; comovement interpretations have endogeneity concerns. Use it to motivate shared-dependency analysis, not a current trading parameter.

E06 - McDonald and Siegel (1986)

The Value of Waiting to Invest. Quarterly Journal of Economics 101(4), 707-727; earlier NBER 1019 (1982). NBER record; author institution.

Inspected: primary abstracts and publication record; full PDF denied. **Supports:** uncertainty and irreversibility can give waiting value beyond a positive-NPV rule. **Limit:** no paper-specific numerical threshold is imported; tradable securities differ from irreversible projects. Waiting must charge competition, quote loss and existing-position risk.

Runtime and distributed systems

O01 - Akidau et al. (2015)

The Dataflow Model. PVLDB. Primary PDF.

Inspected: event/processing time, triggers, windows and watermarks. **Supports:** explicit correctness/latency/cost tradeoffs for unbounded out-of-order streams. **Limit:** watermark progress does not establish a trading fact's prior availability. Corrections cannot retroactively inform an old decision.

O02 - FIX Global Technical Committee (July 2020, EP258)

Order State Changes. Official specification PDF; official landing page.

Inspected: version, cancel, replace and sequencing matrices. **Supports:** pending cancellation/replacement still permits execution; quantities and identifiers matter beyond headline status. **Limit:** the inspected document is dated 2020 despite “latest” in the page title. Verify each actual broker/venue protocol, corrections and reconnect behavior before coding an adapter.

O03 - Featonby, Making Retries Safe with Idempotent APIs

Official AWS Builders' Library article. Publication date not established; accessed October 24, 2026 UTC.

Inspected: tokens, atomic mutation, semantic equivalence and late requests. **Supports:** idempotency is a server/client contract with identity and retention semantics. **Limit:** assigning a UUID locally does not make an arbitrary broker endpoint idempotent. Ambiguous outcomes may require reconciliation rather than new-ID retries.

O04 - Russell and Wefald (1989)

On Optimal Game-Tree Search Using Rational Meta-Reasoning. IJCAI. Primary PDF.

Inspected: value-of-computation, action selection and approximation assumptions. **Supports:** computations can be selected by expected improvement after time/cost. **Limit:** demonstrated game search is not financial alpha; LLM confidence is not measured information value. A one-step approximation may miss useful sequences.

O05 - Dean and Barroso (2013)

The Tail at Scale. Author PDF; author research record.

Inspected: queueing/shared resources and fan-out latency. **Supports:** tail delays can dominate systems waiting for many branches. **Limit:** web-service results are not measured trading latency. Redundant read-request techniques cannot justify duplicating mutating orders. Benchmark the actual end-to-end workload.

O06 - Sculley et al. (2015)

Hidden Technical Debt in Machine Learning Systems. NeurIPS. Primary PDF.

Inspected: data dependencies, entanglement, feedback and configuration. **Supports:** operational complexity extends beyond the fitted model. **Limit:** simplicity does not automatically create profit; compare incremental value against maintenance and failure burden. Explicit ownership and dependencies make changes reviewable.

O07 - Lamport (1978)

Time, Clocks, and the Ordering of Events in a Distributed System. Author-hosted PDF.

Inspected: happened-before, logical clocks and total order. **Supports:** consistent ordering from communication dependencies. **Limit:** logical order does not establish physical-time order in the converse direction. Deterministic replay cannot repair clock error or prove an economic lead-lag relation.

Causal transfer

X01 - Peters, Bühlmann and Meinshausen (2016 publication)

Causal inference by using invariant prediction: identification and confidence intervals. JRSS B 78(5), 947-1012. Author preprint record; inspected PDF.

Inspected: invariance assumption, Proposition 1, Theorem 1, hidden-variable and target-intervention qualifications. Record identifies November 2015 v3; retrieved PDF displays a later compilation date, not treated as a new study. **Supports:** invariance across appropriate environments can restrict causal predictors under structural assumptions. **Limit:** restrictions on interventions affecting the target and confounding matter. Stable backtest bins do not establish causality; an empty identified set can be the honest result.

6 / Validation and use

Completed checks

The updated skill passed its structure validator. Cross-reference and file checks were performed across the original and added material. Independent mathematical and operational review led to explicit hard-fault exits, atomic revalidation before reservation, point-in-time regime fitting, comparable continuation rights in information-value calculations, and clearer no-trade notation.

The dependency-free offline arbiter passed 20 synthetic test methods, including paired robust value versus incorrectly subtracted minima; strict hurdles; deterministic selection; action-specific freshness; expiry; evidence and approval flags; unknown or stale own state; infeasible incumbent recovery; opposing pending orders; simultaneous reservations; joint portfolio stress; signed limits; malformed probabilities/units; partial fills; finite computation caps; and numeric validation. An oversized-integer input was added to the malformed-number test after review. These are invariant checks, not financial backtests.

Two fresh-context applications used only the skill and raw fictional desk problems. One diagnosed unsupported routing uplift, future-fitted regime parameters, invalid selected/reset monitoring, size effects and delayed evidence, then proposed identifiable experiments. The other computed the paired robust difference correctly, rejected unsafe flatten/retry logic, retained pending exposure and removed a two-second optional computation from a 100-millisecond quote path. They produced substantive solutions without being given the intended diagnosis. These are two qualitative cases, not a benchmark or proof of universal LLM behavior.

Checked illustrative arithmetic includes candidate utilities (5,1) versus incumbent (0,4): separate minima differ by +1, while worst paired improvement is -3; and the probability-interval examples producing worst mean improvements 1.95 and 2.35. A possible long inventory of 40-100 followed by an 80-unit sale can become 40 short. No live observations were used to manufacture these examples.

Not established

No strategy backtest on new market data, forward paper-trading record, actual queue-fill experiment, live trade, production adapter, causal uplift estimate or real trading latency benchmark was performed in this addendum. No return multiple or orders-of-magnitude improvement is claimed. Source findings and engineering synthesis are kept distinct. The source register states which paper sections were inspected and which entries had abstract-only access.

The offline kernel consumes supplied models; it does not estimate their correctness. It supports additive scenario constraints, not arbitrary nonlinear margin. It does not implement durable journaling, atomic external reservations, broker deduplication or wall-clock deadlines. The runtime chapter specifies those integration requirements and tests.

Use and promotion

Invoke the saved skill as $engineer-trading-edge and identify the strategy or desk to assess. Preserve its existing baseline and timing authority. The next decisive practical step is to reproduce that baseline on point-in-time data, identify the largest loss of net value, and evaluate one small controller intervention under the same portfolio constraints.

Use the accompanying standalone prompt when the installed skill is unavailable. Keep implementation small until measured evidence justifies a richer model or solver.